Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: 13

Start date: 1st February 2026

Names: Uttam Tripathi

Internship Duration: 8 Weeks

1. Project Title

InfyVision: Infosys Stock Price Prediction Using LSTM Networks

2. Project Objective

Goal: To predict the next-day closing stock price of Infosys (NSE: INFY.NS) using LSTM networks.

Purpose: To provide both an accurate price forecast and a quantified risk assessment, enabling more informed investment decisions.

3. Project description in detail

Approach: Built an end-to-end pipeline involving:

- Data Collection: Fetched 5 years of historical data via Yahoo Finance.
- Data Cleaning: Removed missing or infinite values, flattened multi-index columns, and ensured data integrity to prevent errors during model training.
- Feature Engineering: Created five technical indicators (GARCH, RSI, MACD, Volume Change, Close/Open Ratio).
- Modeling: Trained a stacked LSTM neural network for sequential time-series learning.
- Risk Analysis: Implemented a Monte Carlo simulation to model potential future price paths.
- Deployment: Integrated all components into an interactive Streamlit dashboard.

Technologies Used: Python, TensorFlow/Keras, Pandas, Scikit-learn, Arch, Plotly, Streamlit.

Real-World Impact: The dashboard allows stakeholders to view the current market state, an AI-driven forecast, model performance, and a probabilistic risk outlook (e.g., Value at Risk) for the next 30 trading days.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project Planning & Requirement Analysis	Finalized project scope, defined objectives, and outlined the technical pipeline.
Week 2	Project Setup & Data Collection	Configured environment and fetched 5 years of INFY data.

Week 3	Feature Engineering & Preprocessing	Cleaned data, calculated technical indicators, and created 60-day sequences.
Week 4	Model Architecture & Training	Built and trained the stacked LSTM (64, 32 units).
Week 5	Model Evaluation & Refinement	Evaluated using MAE, MAPE, and Direction Accuracy.
Week 6	Next-Day Prediction & Risk Simulation	Generated next-day price prediction and implemented Monte Carlo simulation (500 runs, 30 days).
Week 7	Dashboard Development & Testing	Built and tested all three dashboard tabs.
Week 8	Documentation, Finalization & Presentation	Completed project report, PowerPoint presentation, final deliverables, and final presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Finalized project scope, defined objectives, and outlined the technical pipeline.	Week 1
Prototype/First Draft	Completed data pipeline, feature engineering, and initial LSTM model training.	Week 4
Mid-Term Review	Generated next-day price prediction and evaluated model using MAE, MAPE, and Direction Accuracy.	Week 6
Final Submission	Integrated Monte Carlo risk simulation (500 runs, 30 days) and completed interactive Streamlit dashboard.	Week 8
Presentation	Delivered final project presentation with live dashboard demonstration.	Week 8

5b. Project execution details

Data Integrity: Built a robust data pipeline with strict chronological splits to prevent data leakage.

Feature Engineering: Transformed raw prices into five meaningful technical indicators for the model.

Model Development: Iteratively designed and trained an LSTM network against a chronological test set until target metrics (MAPE < 3%) were met.

Risk Quantification: Extended the project's scope by implementing a Monte Carlo simulation to provide probabilistic risk metrics (VaR, loss probability).

Deployment: Packaged all components into a cohesive Streamlit dashboard, enabling real-time forecasts and analysis based on the latest market data.

6. Snapshots / Screenshots

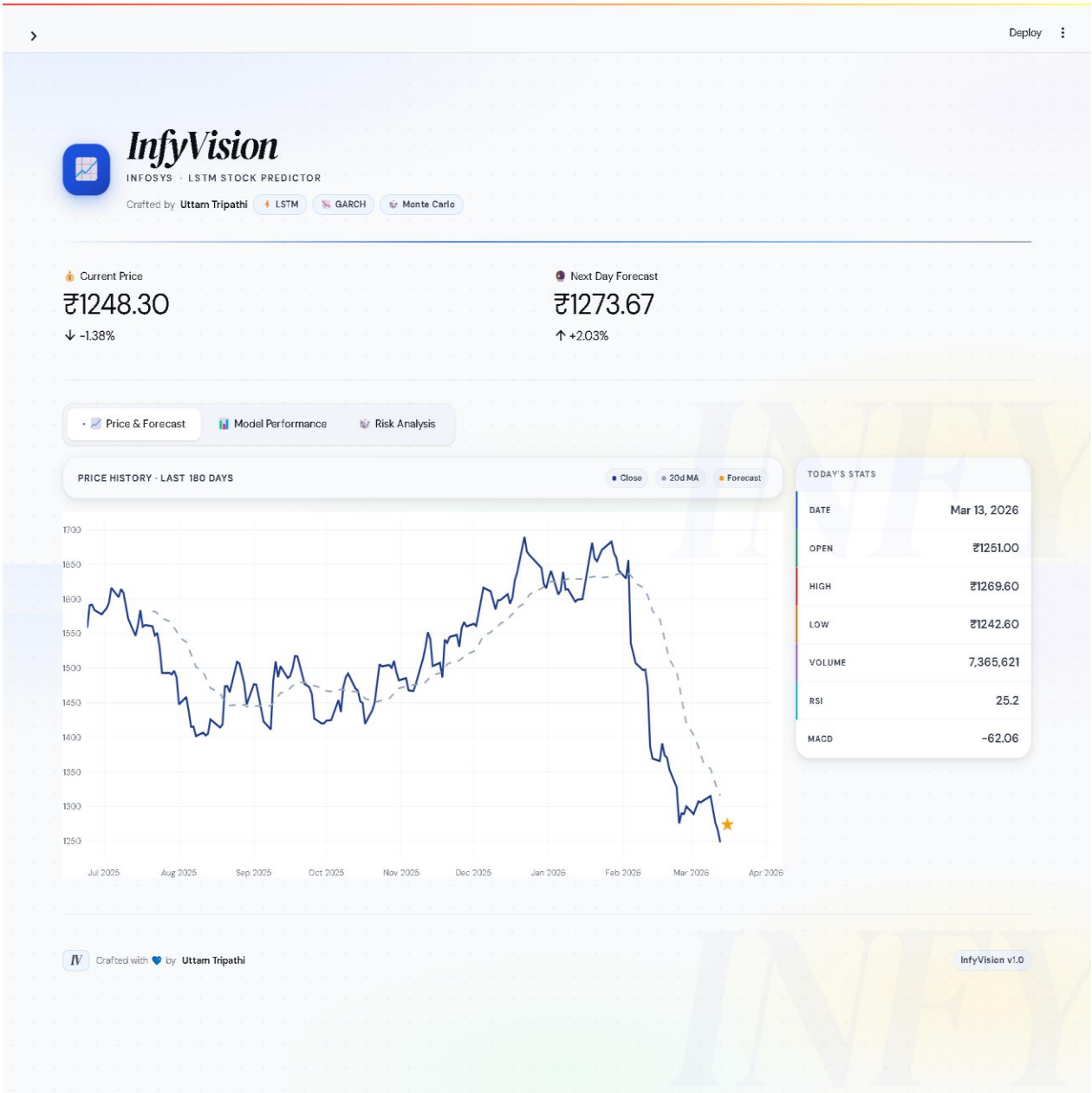

Figure 1: Main dashboard showing price history, 20-day moving average, and next-day forecast.

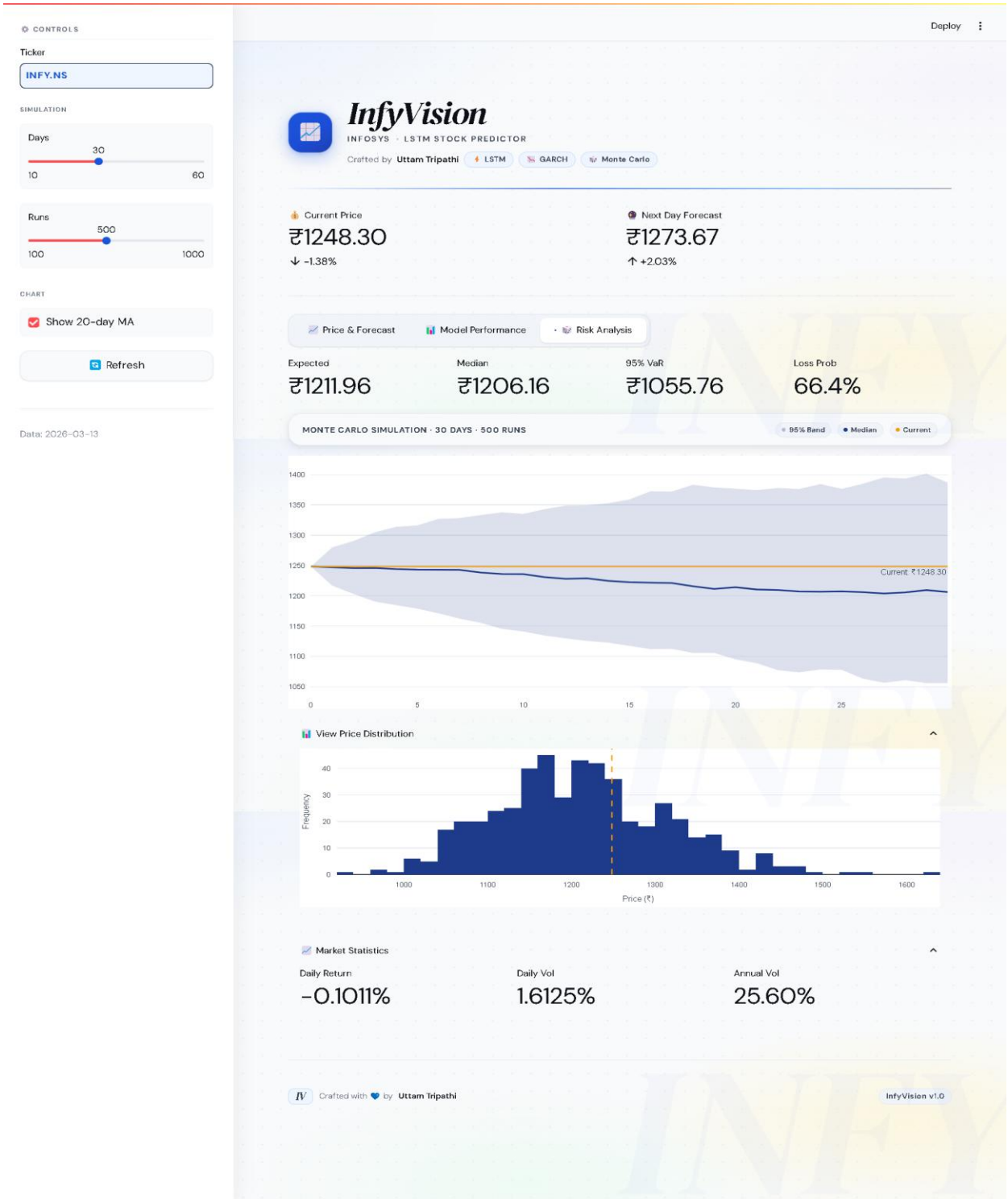

Figure 2: Risk analysis tab showing Monte Carlo simulation results and risk metrics.

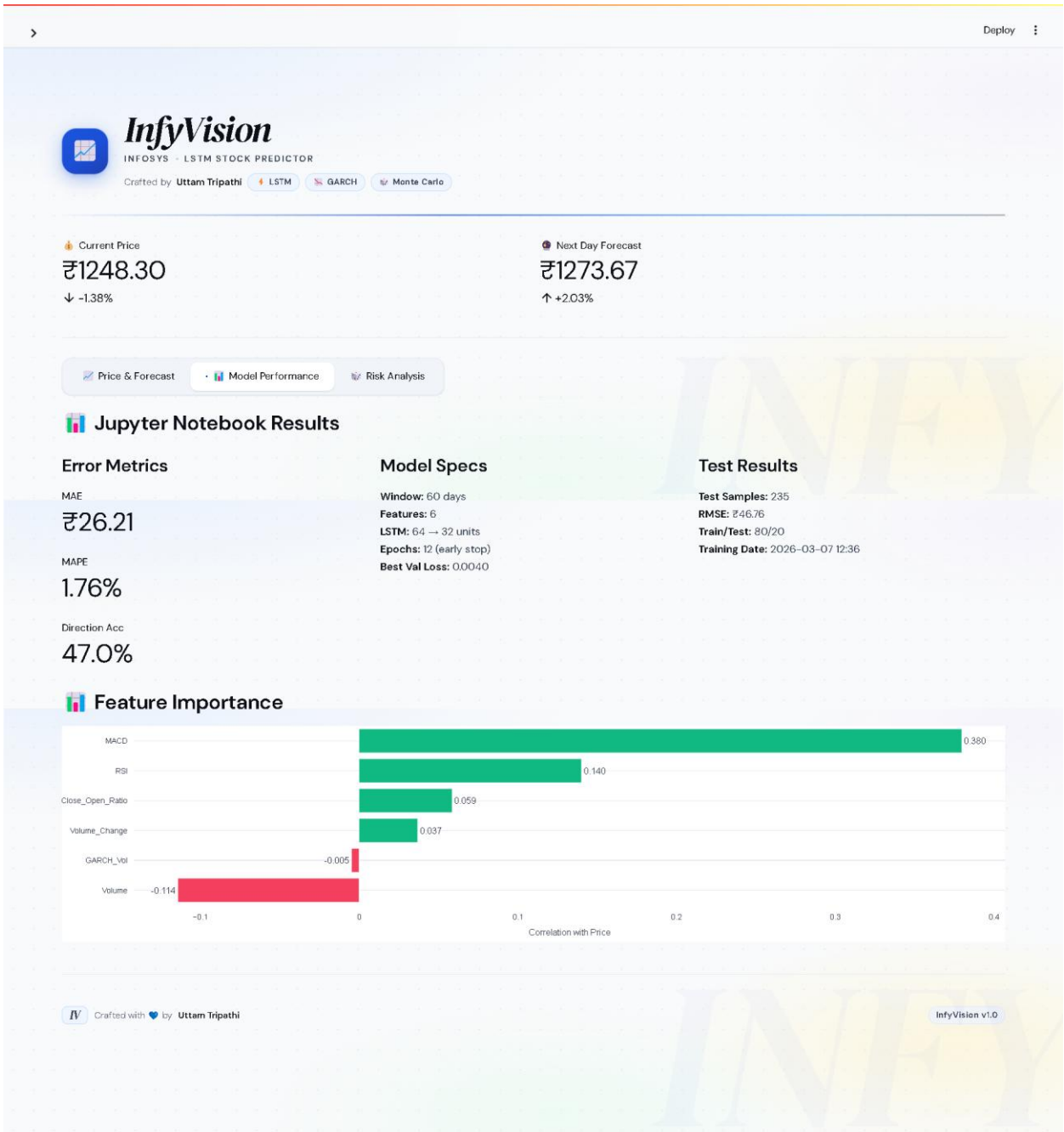

Figure 3: Model performance metrics tab.

7. Challenges Faced

Several challenges were encountered and resolved during the internship:

- Data Leakage: Ensuring the model didn't learn from future data was critical. This was resolved by strictly setting the data end date to "today" and using a strict chronological train/test split

- RSI Calculation Errors: A division-by-zero error occurred during RSI calculation. This was fixed by handling cases where the loss was zero, setting the RSI to 100 for those periods.
- GARCH Model Failure: The GARCH model occasionally failed to converge. A robust fallback using a rolling standard deviation was implemented to ensure the pipeline wouldn't break.
- Model Overfitting: The initial model showed signs of overfitting. This was mitigated by implementing early stopping with patience to preserve the best model weights.

8. Learnings & Skills Acquired

Technical Skills: TensorFlow/Keras for LSTM, Pandas for data manipulation, Plotly for visualizations, and financial modeling (GARCH, Monte Carlo simulations).

Domain Knowledge: Time-series analysis, technical indicators (RSI, MACD), and risk quantification (Value at Risk).

Soft Skills: Project planning, systematic debugging, and translating technical results into user-friendly dashboards.

9. Testimonials from team

Successfully built a fully functional stock prediction system with integrated risk analysis.

Achieved the transition from a Jupyter notebook prototype to a polished, interactive Streamlit app.

Solidified understanding of deep learning's practical applications in finance.

Demonstrated the power of combining AI with traditional financial modeling to create a tool with real-world utility.

10. Conclusion

InfyVision internship successfully delivered a production-ready stock prediction system.

The project met its primary technical objectives, including achieving a low percentage error (MAPE < 3%).

It successfully integrated comprehensive risk quantification (VaR, loss probability).

This experience has significantly strengthened my skills in applied AI and financial analytics, directly aligning with my career goals in data science and machine learning engineering.

11. Acknowledgements

I extend my sincere gratitude to my mentor, Keerthana, for her invaluable guidance and support throughout this project.

I also thank the Infosys Springboard program for providing this excellent platform for learning and practical application.